Constrained sphere packing and the interpolation problem for positive definite functions

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Joint work with Sujit Sakharam Damase, IISc, Bangalore

叫我老潘，就像那个法国角色一样。

The team 高山流水

他即是数学力量的化身，法力无边。

Acknowledgments 饮水思源

We would like to thank:

- The Indian Institute of Science, Bangalore
- The Indian Statistical Institute, Bangalore
- J C Bose Fellowship JCB/2021/000041 of SERB

for supporting this research.

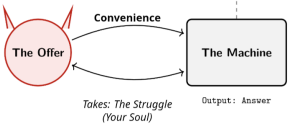
听听我的《西游记》

The future is now

Algebra is the offer made by the devil to the mathematician. The devil says: I will give you this powerful machine, it will answer any question you like. All you need to do is give me your soul: give up geometry and you will have this marvelous machine.

—*Sir Michael Atiyah*, 2002

$P(w_t|w_{<t}) = \text{softmax}(h_t)$



AI is the offer made by the devil to the mind.

The devil says: I will give you this powerful machine, it will complete any thought you like.

All you need to do is give me your soul: give up ***the struggle*** and you will have this marvelous machine.

— Gemini, 2025

"I 爱 AI": Powered by **Gemini** & **DeepSeek** 鹊桥相会

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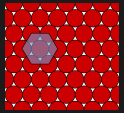
Lagrange



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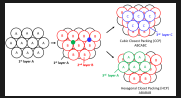
Hales (90s)



n=8,24
Viazovska et al. (2010s)

E_8,
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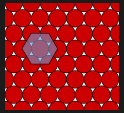

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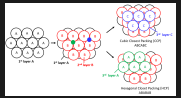

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For no other cases is the optimal density known exactly, although the optimal packings do not come from lattices.

Spherical codes

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A **spherical code** is a set of points in the the sphere in n real dimensional space. We say it has **minimum angle** to be the smallest θ such that every pair of distinct points in the code makes an angle of at least θ with respect to the origin.

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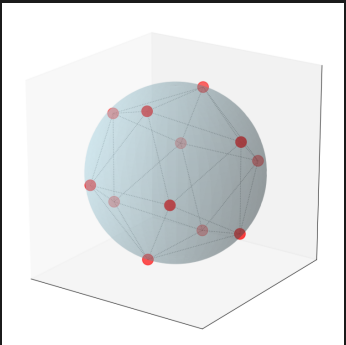

Kissing number problem 亲密无间

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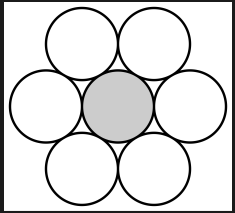

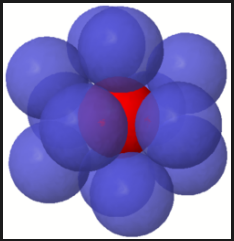

The kissing number is equal to the maximal size of a spherical code with minimum angle $\pi/3$.

Delsarte's method 画地为牢

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Strategy: Given $n \in \mathbb{N}$ and $0 < \theta < \pi/2$ find the maximum $\tilde{\theta} > \pi/2$ such that there is a map from the sphere in n-dimensional space to the sphere in a larger dimensional space such that every spherical code with minimum angle θ is mapped to a code with minimum angle at least $\tilde{\theta}$.

Positive definite functions

Positive definite functions

Let G be group, let H be subgroup of G. A (completely) positive definite function

$$f : G/H \to \mathbb{R} \text{ (resp. self-adjoint operators)}$$

satisfies: $[f(g_i^* g_j)]_{i,j}$ is positive semidefinite for all $g_1, \ldots, g_d \in G$.

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$$f(g) = v^* \pi(g) v$$

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The homomorphic analogue of **Tomiyama's theorem** gives that f is in fact bi-invariant,

$$f(h_1 g h_2) = f(g), \forall h_1, h_2 \in H, g \in G.$$

Schoenberg's theorem on positive definite functions

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If f is positive definite and continuous, $f(g) = \sum_{i=1}^{\infty} a_i C_i^{(n)}(\gamma_{11})$ where $a_i \geq 0$ and the $C_i^{(n)}$ are Gegenbauer polynomials, a natural basis for bi-invariant functions. (We will normalize to have $C_i^{(n)}(1) = 1$) (Schoenberg '42, Bochner '41)

Operations on positive definite functions

Operations on positive definite functions

Given $C_i^{(n)}(\gamma_{11}) = v_i^* \pi_i(\gamma) v_i$, we see that

$$C_i^{(n)} C_j^{(n)} = \sum_{k=0}^{i+j} a_k C_k^{(n)}$$

where $\sum a_k = 1, a_k \geq 0$ as

$$C_i^{(n)}(\gamma_{11}) C_j^{(n)}(\gamma_{11}) = (v_i \otimes v_j)^* (\pi_i \otimes \pi_j)(\gamma)(v_i \otimes v_j)^*$$

and $\pi_i \otimes \pi_j$ may again be broken into irreducibles.

Kernel embedding 别有洞天

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Theorem (Aronszajn's theorem)

Let X be a set. Let $A : X \times X \to \mathbb{C}$ such that $[A(x_i, x_j)]_{i,j}$ is positive semidefinite for arbitrary finite collections $x_1, \ldots, x_d$.
There exists a Hilbert space $\mathcal{H}$ and $\iota_x : X \to \mathcal{H}$ such that $\langle \iota_x, \iota_y \rangle = A(x, y)$.

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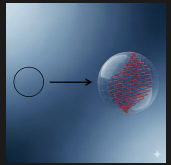
Thus a positive definite function $f : [-1, 1] \to \mathbb{R}$ such that $f(1) = 1$ induces an embedding via the kernel $f(\langle x, y \rangle)$ which lands in the sphere of the Hilbert space as $f(\langle x, x \rangle) = 1$.



Delsarte's method 神来之笔

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Theorem (Delsarte '72)

Let $0 < \theta < \pi$. Let $g : [-1, 1] \to \mathbb{R}$ be a positive definite function on the sphere in n dimensions such that $g(1) = 1$ and $g(x) \leq 0$ if $x \leq \cos\theta$.
Let $\bar{g}$ the constant coefficient of g in its Gegenbauer expansion. (That is the average value of $g(\langle x, y\rangle)$ over x, y in the unit sphere. Let $\mathcal{C}$ be spherical code with minimal angle at most θ. Then,

$$|\mathcal{C}| \leq \frac{1}{\bar{g}}.$$

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Linear programming methods: Delsarte-Goethals-Seidel, '77 Kabitiansky-Levenshtein '78.

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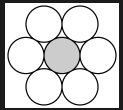

The polynomial

$$p(t) = \frac{4}{9}(t - 1/2)(t + 1/2)^2(1 + t)$$

satisfies

$$p(x) = \frac{1}{6}C_0^{(2)}(x) + \frac{1}{3}C_1^{(2)}(x) + \frac{5}{18}C_2^{(2)}(x) + \frac{1}{6}C_3^{(2)}(x) + \frac{1}{18}C_4^{(2)}(x)$$

$$\text{i.e. } p(\cos\theta) = \frac{1}{6} + \frac{1}{3}\cos\theta + \frac{5}{18}\cos 2\theta + \frac{1}{6}\cos 3\theta + \frac{1}{18}\cos 4\theta$$

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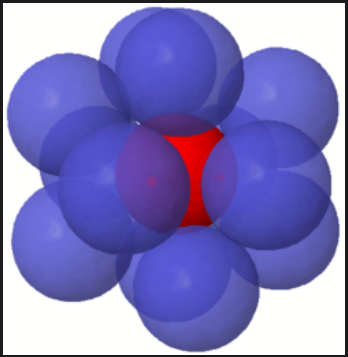

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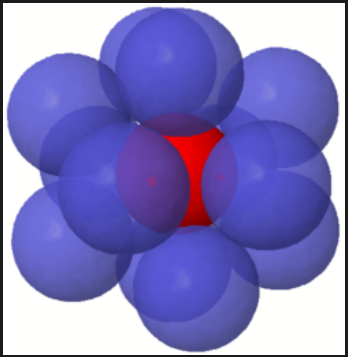

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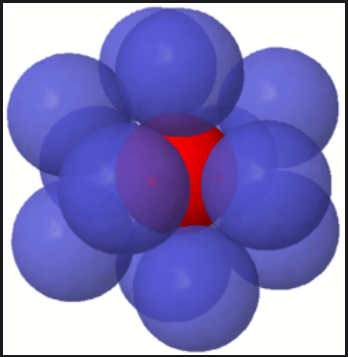

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Sharp	Not Sharp
$n = 2$ kissing number	$n = 3, 4$ kissing number
	Musin
$n = 8, 24$ kissing number	$X = \{-1, 1/2, -1/2, 1\}, n = 10$
Sloane, Odlyzko, 79	Damase, P. ($\frac{1}{\bar{g}} = 46.$)

Cohn-Elkies method 如来神掌

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Theorem (Cohn-Elkies, '03)

Suppose $f : \mathbb{R}^n \to \mathbb{R}$ is an admissible function, is not identically zero, and satisfies the following two conditions:

1. $f(x) \leq 0$ *for* $|x| \geq 1$, *and*
2. $\hat{f}(t) \geq 0$ *for all* t.

Then the center density of n-dimensional sphere packings is bounded above by

$$\frac{f(0)}{2^n \hat{f}(0)}.$$

(从天而降的掌法)

Constrained angle codes

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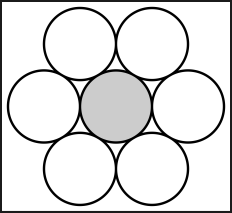

For example, the spherical code given by six equispaced points on the circle has cosines in $\{0, -1, 1/2, -1/2\}$.

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When is there a positive definite function on the sphere in n-dimensions $\tilde{f} : [-1, 1] \to \mathbb{R}$ such that $\tilde{f}|_X = f$?

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is a map which naturally induces a completely positive map on the operator system corresponding to Γ inside the group algebra $\mathbb{C}[G]$. c.f. Krein, Jorgensen, etc. on similar but different interpolation problems.

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Theorem (Damase-P.)

$SO(\infty)/1 \oplus SO(\infty)$ and $SO(2)/\{1\}$ are not pathological.

Continuous extension problem 大道至简

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Let X be a finite subset of $[-1, 1]$. Let $f : X \to \mathbb{R}$. When is there a continuous positive definite function on the sphere in n-dimensions $\tilde{f} : [-1, 1] \to \mathbb{R}$ such that $\tilde{f}|_X = f$?

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Theorem (Damase, P.)

Let f correspond to a solvable continuous extension problem. There is a polynomial solution $\tilde{f}$.

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Viazovska et al. functions for Cohn-Elkies are modular functions. (Low algebraic complexity.)

Delsarte hallucinations 镜花水月

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- Consider $X = \{-1, 1/2, -1/2, 1\}, n = 10$. There is no corresponding constrained angle code, but we have $\frac{1}{\bar{g}} = 46$ is an integer. Many such examples exist of **Delsarte hallucinations.**

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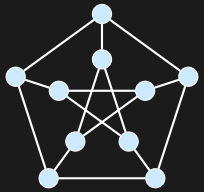
- On the other hand, the example $X = \{1/3, -1/3, 1\}$ has a spherical code with 10 points in the 5 dimensional sphere, but the sign pattern comes from a Petersen graph.
- Given an X, determining if there exists a subset of the n-sphere with cosines in X of some size k is a difficult combinatorial matrix completion problem.

Delsarte Vision 水落石出



The Petersen Graph

Seidel Matrix S

$$\begin{pmatrix} 0 & - & - & 1 & 1 & - & 1 & 1 & 1 & 1 \\ - & 0 & - & - & 1 & 1 & - & 1 & 1 & 1 \\ - & - & 0 & - & - & 1 & 1 & - & 1 & 1 \\ 1 & - & - & 0 & - & 1 & 1 & 1 & - & 1 \\ 1 & 1 & - & - & 0 & 1 & 1 & 1 & 1 & - \\ - & 1 & 1 & 1 & 1 & 0 & 1 & - & 1 & - \\ 1 & - & 1 & 1 & 1 & 1 & 0 & 1 & - & - \\ 1 & 1 & - & 1 & 1 & - & 1 & 0 & - & - \\ 1 & 1 & 1 & - & 1 & 1 & - & - & 0 & - \\ 1 & 1 & 1 & 1 & - & - & - & - & - & 0 \end{pmatrix}$$

Spectrum: $3^{(5)}, -3^{(5)}$

Gram Matrix G

Construct $X = \{1, \pm 1/3\}$:

$$G = I + \frac{1}{3} S \succeq 0$$

Discovery: AI = 爱

- Example suggested by **Gemini**.
- Verified & Formalized by **Damase & P.**

登峰造极

让我们发展数学，直到它充满整个天空。

满载而归

Constrained sphere packing and the interpolation problem for positive definite functions

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叫我老潘..... 就像那个法国怪盗一样。